EEK 2BIS.ipynb ☆

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```
================================================================================
ALL ROUTING PATHWAY SYSTEMS WATER FLOW REPORT
================================================================================

❌ NON-OPTIMAL —> Layout 1
--------------------------------------------------------------------------------
  DAM ═══> Area_A   | Sent: 300.0L | Delivered: 300.0L | Lost:  0.0L
  DAM ═══> Area_B   | Sent: 529.4L | Delivered: 450.0L | Lost: 79.4L
  DAM ═══> Area_C   | Sent: 277.8L | Delivered: 250.0L | Lost: 27.8L
  DAM ═══> Area_D   | Sent: 150.0L | Delivered: 150.0L | Lost:  0.0L
  Result Status: Total Pipeline Loss = 107.2 Liters.
--------------------------------------------------------------------------------

❌ NON-OPTIMAL —> Layout 2
--------------------------------------------------------------------------------
  DAM ═══> Area_A   | Sent: 333.3L | Delivered: 300.0L | Lost: 33.3L
  DAM ═══> Area_B   | Sent: 450.0L | Delivered: 450.0L | Lost:  0.0L
  DAM ═══> Area_C   | Sent: 312.5L | Delivered: 250.0L | Lost: 62.5L
  DAM ═══> Area_D   | Sent: 157.9L | Delivered: 150.0L | Lost:  7.9L
  Result Status: Total Pipeline Loss = 103.7 Liters.
--------------------------------------------------------------------------------

🏆 OPTIMAL —> Layout 3
--------------------------------------------------------------------------------
  DAM ═══> Area_A   | Sent: 300.0L | Delivered: 300.0L | Lost:  0.0L
  DAM ═══> Area_B   | Sent: 450.0L | Delivered: 450.0L | Lost:  0.0L
  DAM ═══> Area_C   | Sent: 250.0L | Delivered: 250.0L | Lost:  0.0L
  DAM ═══> Area_D   | Sent: 150.0L | Delivered: 150.0L | Lost:  0.0L
  Result Status: Total Pipeline Loss = 0.0 Liters.
--------------------------------------------------------------------------------
```

WEEK 2BIS.ipynb

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[1] ✓ 0s

```
def get_flows(name):
    flows = {k: [DEMANDS[k] / eff, (DEMANDS[k] / eff) - DEMANDS[k]] for k, eff in LAYOUTS[name].items()}
    return sum(f[1] for f in flows.values()), flows

fit = lambda ind: -get_flows(NAMES[int(ind)])[0]
pop = np.random.randint(0, len(NAMES), size=(30, 1))


for _ in range(50):
    scores = np.array([fit(ind) for ind in pop])
    best = pop[np.argmax(scores)].copy()

    next_gen = [best, best]
    while len(next_gen) < 30:
        p1, p2 = pop[np.random.choice(30, 2)]
        c1 = p2 if np.random.rand() < 0.8 else p1
        if np.random.rand() < 0.2: c1 = [np.random.randint(0, len(NAMES))]
        next_gen.append(np.array(c1))
    pop = np.array(next_gen[:30])


print("="*85 + "\nALL ROUTING PATHWAY SYSTEMS WATER FLOW REPORT\n" + "="*85)
for name in NAMES:
    loss, flows = get_flows(name)
    tag = "🏆 OPTIMAL" if name == NAMES[int(best[0])] else "❌ NON-OPTIMAL"
    print(f"\n{tag} —> {name}\n" + "-"*85)
    for k, f in flows.items():
        print(f"  DAM ——> {k:<8} | Sent: {f[0]:>5.1f}L | Delivered: {DEMANDS[k]:>5.1f}L | Lost: {f[1]:>4.1f}L")
    print(f"  Result Status: Total Pipeline Loss = {loss:.1f} Liters.\n" + "-"*85)
```

[1]
✓ 0s

```python
import numpy as np


DEMANDS = {'Area_A': 300, 'Area_B': 450, 'Area_C': 250, 'Area_D': 150}
LAYOUTS = {
    'Layout 1 ':  {'Area_A': 1.00, 'Area_B': 0.85, 'Area_C': 0.90, 'Area_D': 1.00},
    'Layout 2 ': {'Area_A': 0.90, 'Area_B': 1.00, 'Area_C': 0.80, 'Area_D': 0.95},
    'Layout 3 ': {'Area_A': 1.00, 'Area_B': 1.00, 'Area_C': 1.00, 'Area_D': 1.00}
}
NAMES = list(LAYOUTS.keys())


def get_flows(name):
    flows = {k: [DEMANDS[k] / eff, (DEMANDS[k] / eff) - DEMANDS[k]] for k, eff in LAYOUTS[name].items()}
    return sum(f[1] for f in flows.values()), flows

fit = lambda ind: -get_flows(NAMES[int(ind)])[0]
pop = np.random.randint(0, len(NAMES), size=(30, 1))


for _ in range(50):
    scores = np.array([fit(ind) for ind in pop])
    best = pop[np.argmax(scores)].copy()

    next_gen = [best, best]
    while len(next_gen) < 30:
        p1, p2 = pop[np.random.choice(30, 2)]
        c1 = p2 if np.random.rand() < 0.8 else p1
```